In [1]:
```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, confusion_matrix
```

In [2]:
```
data = pd.read_csv("The Titanic dataset.csv", header=1)
```

In [3]:
```
data.head()
```

Out[3]:

	sn	pclass	survived	Unnamed: 3	gender	age	family	fare	embarked	date
0	1	3	0	Mr. Anthony	male	42	0.0	7.55	NaN	01-Jan-90
1	1	3	0	Mr. Anthony	male	42	0.0	7.55	NaN	01-Jan-90
2	2	3	0	Master. Eugene Joseph	male	?	2.0	20.25	S	02-Jan-90
3	3	2	0	Abbott, Mr. Rossmore Edward	NaN	NaN	2.0	**	S	03-Jan-90
4	4	3	1	Abbott, Mr. Rossmore Edward	female	35	2.0	20.25	S	04-Jan-90

In [4]:
```
data.replace('?', pd.NA, inplace=True)
```

In [5]:
```
# Convert columns to numeric
data['pclass'] = pd.to_numeric(data['pclass'], errors='coerce')
data['age'] = pd.to_numeric(data['age'], errors='coerce')
data['fare'] = pd.to_numeric(data['fare'], errors='coerce')
data['survived'] = pd.to_numeric(data['survived'], errors='coerce')
```

In [6]:
```
# Convert gender to numeric
data['gender'] = data['gender'].map({'male': 0, 'female': 1})
```

In [7]:
```
# Fill missing values
data.fillna(0, inplace=True)
```

In [8]:
```
X = data[['pclass', 'gender', 'age', 'fare']]
y = data['survived']
```

In [9]:
```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

In [10]:
```
model = LogisticRegression(max_iter=1000)
model.fit(X_train, y_train)
```

Out[10]:
```
LogisticRegression
LogisticRegression(max_iter=1000)
```

In [11]:
```
y_pred = model.predict(X_test)
```

In [12]:
```
accuracy = accuracy_score(y_test, y_pred)
cm = confusion_matrix(y_test, y_pred)
```

In [13]:
```
print("Accuracy:", accuracy)
print("Confusion Matrix:\n", cm)
```

```
Accuracy: 0.7854406130268199
Confusion Matrix:
 [[122  27]
 [ 29  83]]
```

In [ ]: